Trends in
Ecology & Evolution

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Opinion

From rivals to partners: adversarial collaboration in ecology and evolution

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Since their inception, ecology and evolutionary biology have been filled with controversies. While controversies are generally beneficial, they can sometimes hinder meaningful communication and, ultimately, decelerate scientific advancement. A clear example is when scientific rivals enter a point-scoring mode in which proving the opponent wrong sidelines objective assessments. To counter this, we introduce to the ecology and evolutionary biology audiences the adversarial collaboration approach, where opponents cooperatively address their disagreements to advance their field. We describe the logic of adversarial collaboration, examine a series of potential challenges to its successful adoption in these disciplines, and show how they can be overcome. Adversarial collaboration is an invitation to align controversies with the scientific ethos of collectively seeking empirical truths.

Highlights

Scientific controversies can sometimes lead to confrontational interactions in which meaningful communication between rivals breaks down.

Adversarial collaboration proposes that rivals cooperate in a structured manner to tackle their disagreements.

Adversarial collaboration has not been widely applied in ecology and evolution, partly due to a series of potential hindrances in these disciplines.

We address these challenges, suggesting ways to adapt adversarial collaboration to the specific features of controversies in ecology and evolution.

Controversies: the good and the bad

Scientific controversies arise when scientists working in the same field or on similar problems disagree extensively on key issues of their discipline [1,2]. In ecology and evolutionary biology, examples include debates over the role of neutral processes in evolution [3,4], whether genetic or demographic factors drive species extinction [5], the explanatory power of Hamilton's rule [6], or the need for an extended evolutionary synthesis [7–11].

Controversies are normal and have positive consequences for science, helping sharpen arguments, spurring theory development, and inspiring innovative empirical research [12]. However, they can sometimes get entrenched, potentially slowing down scientific progress. In particular, the lack of agreement over data, methods, and conceptual frameworks can favor what Daniel Kahneman called 'angry science' [13], a confrontational approach to debates where scoring points against the opponent takes precedence over impartial evaluations. Angry science can manifest in cycles of replies and rejoinders to adversaries' papers, which easily degenerate into unproductive fights where meaningful communication breaks down [13]. These communication failures can prevent the gathering of important data or the clarification of theoretical disagreements, two important preconditions for scientific progress.

One possible way to avoid this situation, keeping controversies constructive and productive, is an approach known as adversarial collaboration, or cooperation between opponents to bring progress into their disagreement. Although there has recently been a growing interest in adversarial collaboration [14–16], it is yet to be adopted widely in ecology and evolution. This may be due to a series of challenges that may seem to limit its implementation in these fields and that are related to the diversity of organisms studied, the multiplicity of processes that contribute to ecological and evolutionary change, and to the fact that many controversies in ecology and evolutionary biology involve conceptual rather than factual disagreements. Here, we introduce adversarial

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collaboration and discuss these challenges, showing how they can be overcome so that the ecology and evolutionary biology communities have another tool to help scientific controversies move forward.

What is adversarial collaboration?

Adversarial collaboration consists of a temporary partnership between disagreeing scientists, who, with the help of an arbiter, agree to cooperatively examine their differences and, if possible, advance towards their resolution [14,15,17]. A standard adversarial collaboration would normally be prompted by an adversary's publication on a topic of mutual interest. If such a publication contradicts one's preferred hypothesis or approach, we might feel tempted to write a critical reply, sometimes letting our passions take the lead over impersonal reasons and evidence. What the adversarial collaboration approach recommends instead is reaching out to the corresponding author to offer a collaboration. Such collaboration would be guided by an impartial arbiter, who would be jointly selected from reputed scholars in the field and whose main aim would be to mediate in potential conflicts and ensure that the interaction remains constructive.

To illustrate how adversarial collaboration works, let us focus on empirical controversies, such as whether or not we are currently witnessing a sixth mass extinction [18–20] (for practical guidelines, see also Table 1 in [21]). First, adversaries should discuss their points of disagreement in a way that both sides feel fairly portrayed, agreeing on the hypothesis each side espouses. Competing hypotheses should be formulated as clearly as possible, permitting the deduction of empirically testable predictions that can a priori discriminate between them. Specifically, each participant should clarify beforehand which results would run against their hypothesis, proposing a series of possible interpretations for such dissenting outcomes – the expected results could even be pre-registered. To test these predictions, adversaries must agree upon a study design, specifying as many details of the study as possible, including procedures for data collection and analysis (thus helping minimize analytical flexibility; e.g., [22]). The arbiter should keep a record of all stages of the process to prevent future disagreements prompted by memory lapses or shifts of opinion.

Once the joint study has been designed, participants should carry it out swiftly, so that discord or a decrease in interest is prevented. All parties should commit in advance to coauthor a publication based on the results obtained, whatever these turn out to be. The data collection and analysis should be conducted cooperatively following the agreed protocol, and the study should comply with Open Science practices. Finally, participants should write the manuscript for publication. Preferably, the arbiter would lead the writing of the introduction and the results sections, framing the question, alternative hypotheses, and obtained results as neutrally as possible. The adversaries, for their part, would be responsible for writing the methods and discussion – ideally in collaboration and following strict deadlines. However, if disagreements persist over the interpretation of results, the arbiter would allocate each opponent a predetermined space in the discussion to present their respective views. The decision regarding which journal to submit the manuscript to would also rest in the arbiter's hands, informed by input from the adversaries. See Box 1 for a short history of the concept of adversarial collaboration.

Adversarial collaboration in ecology and evolution: challenges and prospects

At first sight, ecology and evolutionary biology may appear poorly suited for adversarial collaboration. Particularly, researchers might be concerned about the problems posed by the diversity of organisms studied, the multiple processes that underlie ecological and evolutionary change (or lack thereof), and the conceptual nature of many disagreements.

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Box 1. A short history of adversarial collaboration

The idea of adversarial collaboration originated in psychology [23,24]. It gained prominence with a collaboration between Daniel Kahneman and Ralph Hertwig, in which Barbara Mellers acted as arbiter [21]. The controversy revolved around the effect of frequency formats and semantic ambiguities on what psychologists call the conjunction effect. While the interpretation of the results from the experiments was not straightforward, leading adversaries to disagree on their implications, the effort paid off not only in that it generated new relevant evidence but also in that both researchers came to value aspects of their rivals' position.

Since Mellers *et al. [*21], adversarial collaboration has mostly been implemented in psychology and neuroscience (for a summary of the most prominent studies up to 2020, see [17]), not only through new cases [25–27], but also through discussions about how to improve the approach [14–16,25,28–30]. Participants in adversarial collaborations report that, although it requires more effort, it normally leads to more robust results and richer post-publication discussions [31].

Diversity of organisms studied

Unlike psychology, which normally deals with just one species (e.g., humans, or even WEIRD humans, [32]), and many other branches of biological science, where most research is conducted on highly standardized model organisms, ecologists and evolutionary biologists quite literally study all life [33,34]. Successful research on non-model organisms requires detailed knowledge of their natural history – expertise normally possessed only by those who have worked extensively on these species. A plausible expectation is that adversarial collaboration works best when the participants involved are epistemic peers – that is, researchers with reasonably equal skills and knowledge of the problem and the field [35,36]. If so, choosing only one adversary's study system could introduce asymmetries between opponents, potentially compromising the neutrality of the resulting investigation.

To solve this, one possibility is for adversaries to test their competing hypotheses using a so-to-speak 'neutral' species (or a set of neutral species) that neither of them had studied in the past, so that epistemic peerhood is ensured (examples of similar strategies include [37] for species and [38] for populations). Given the required expertise about a species' natural history, this strategy will likely require involving further partners in the collaboration, who would be experts in the chosen organisms. This would certainly make the adversarial collaboration more burdensome. But it can also have advantages if these partners remain neutral with respect to the hypotheses being tested. This can be achieved by recruiting researchers working on a different topic, or even in another biological discipline. These additional partners could reduce frictions and add an extra layer of impartiality to the collaboration.

Alternatively, each participant may design a separate study on their system organism, with rivals committing to provide continuous constructive feedback in advance (similar to Mellers *et al*.'s second and third experiments; [21]). The opponents and moderator would all participate in data collection and analyses, with each rival taking the lead in their respective study. Since both opponents will go through leader and subordinate roles, this approach also ensures epistemic peerhood, although in a sequential manner. In this case, the resulting paper would comprise (at least) two independent studies aimed at testing the same competing hypotheses in two or more different organisms but designed in a way that ensures both have a fair shot of proving true.

Multiplicity of processes

A second reason to doubt the efficacy of adversarial collaboration in ecology and evolutionary biology is that discrepancies commonly revolve not around the specific cause, mechanism, or process responsible for a phenomenon, but over the relative frequency or importance of several possibilities [39,40]. This applies to, for example, disagreements over the role of neutral processes in molecular evolution and ecology [4], the importance of coding versus regulatory change in adaptive evolution [41,42], and whether species go extinct because of inbreeding depression

[5]. Relative frequency debates justify continuing to adhere to a hypothesis after obtaining a negative result, which does little to ease the controversy. For example, a supporter of *cis*-regulatory evolution could maintain their confidence in its explanatory power despite yet another example of coding change, provided that other instances do exist [42].

It is not just how often something happens that is affected by the causal complexity of ecological and evolutionary phenomena, but also the relative explanatory importance or efficacy of specific causes [40]. On many occasions, alternative processes may simultaneously contribute to ecological or evolutionary phenomena, so contrasting hypotheses may differ only in the explanatory weight ascribed to different factors. Because of this, often adversarial collaborators must be content to show that one of the competing factors has a lower explanatory power than previously assumed, but not that it is not important at all. For example, if a study finds that ecological variables can explain a significant amount of variance in brain regions related to cognitive abilities in primates, this does not necessarily entail that social variables are not important [43]. What is more, in many cases, controversies simultaneously revolve around both the frequency and efficacy of different causes [40].

These examples illustrate that the falsificationist approach to adversarial collaboration, aimed at refuting one of the competing hypotheses, is not suitable for ecology and evolutionary biology (see also [30]). A more fitting perspective is whether the tested hypotheses or research programs are progressive or regressive, that is, successfully or unsuccessfully explain new empirical data [44]. In practice, this can involve, for example, determining the direction and effect size of various predictors as outlined in competing hypotheses (e.g., [25]).

While this implies that adversarial collaboration would rarely rely on definite tests, it would also make room for collaborations whose target is not to gather further facts but to assess the general status of competing mechanisms through meta-analyses on published data. Meta-analyses are common in ecology and evolution, but they do not typically involve formal associations between adversaries. Adversarial collaborations can make it more likely that results are considered relevant by all opposing camps. For example, researchers holding opposite views about the role of color polymorphism in species persistence might devise a meta-analytic protocol that can tilt the balance in favor of each of the competing hypotheses [45–48]. Adversarial collaboration could also be applied to theoretical biology. For example, disagreeing peers might decide to launch an adversarial collaboration to assess through mathematical modeling whether monogamy is a strict precondition for the evolution of eusociality [49–53]. Choice prediction competitions in decision science might provide useful guidelines to initiate modeling adversarial collaborations in biology (e.g., [54]).

The conceptual nature of many controversies

The last and perhaps most problematic challenge for adversarial collaboration in ecology and evolution is the conceptual nature of many controversies. For example, debates over the role of development in evolution revolve around what counts as a legitimate cause for adaptive change [55], and the controversy over the neutral theory of biodiversity reflects opposing views on expected or desirable features of models and theories [56]. Conceptual disagreement can make the collaboration between adversaries more complicated, hindering consensus on even things as basic as problem formulation. In these situations, then, the aims of adversarial collaboration must shift from empirical advancement to clarifying disagreements, finding common theoretical ground, and bringing awareness of the scientific legitimacy of the rivals' proposals (i.e., what Jonathan Birch [57] has called 'meta-consensus'). These outcomes are difficult to attain when exchanges of opinion are confrontational rather than collaborative.

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We suggest that progress in conceptual disputes requires paying special attention to three potential meta-scientific domains of disagreement: ontological, epistemological, and semantic [58] (see Eigenbrode *et al*. for a similar approach in the context of interdisciplinary science, [59]). Ontological disagreements are those that researchers have over the basic stuff that populates the biological world, as well as its most fundamental characteristics. These include discrepancies about, for example, the essential properties of organisms and the nature of causation: are organisms vehicles for genes, or goal-directed, self-determining (i.e., agentic) entities [60,61]? Epistemological disagreements relate to differences about what the most important attributes of good science are. For example, what is more important to assess theories: simplicity and mathematical precision, factual accuracy (i.e., relying upon as few idealizations of phenomena as possible), or the degree of integration with other disciplines (e.g., [62])? Finally, semantic disagreements have to do with how different ecologists and evolutionary biologists define essential concepts: for example, what do we mean by 'development' or by 'evolutionary process' [63,64]?

Although these three meta-scientific domains are to some extent responsive to empirical evidence, their connection to facts is rather indirect. This is why it is unlikely that debates

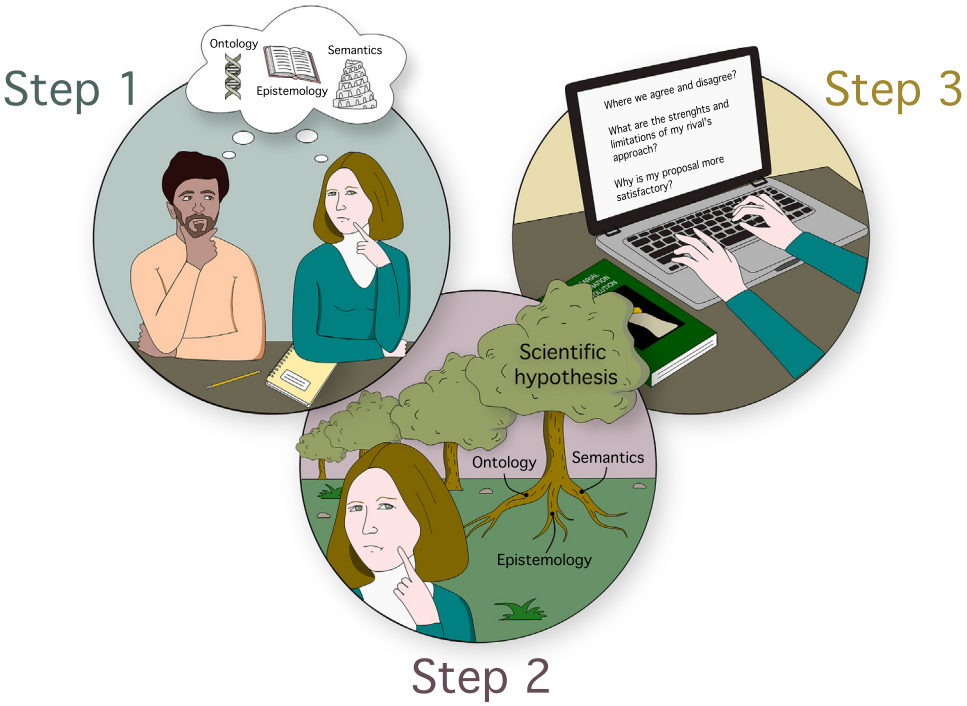


Figure 1. A three-step model of theoretical adversarial collaboration. Step 1 involves rivals jointly reflecting on the ontological, epistemological, and semantic views they hold regarding the topic of dispute, with strict time or space limits set by the arbiter. These reflections may be stimulated by questions (e.g., ontology: 'I understand the central entities and processes on which my theory focuses to be … [e.g., genes, organisms, ecosystems, evolutionary change] and to have such-and-such properties, based on the following evidence and reasons'; epistemology: 'among the attributes that make a scientific theory worthy, I believe … [e.g., simplicity, fruitfulness, integration, mathematical precision] are the most important, for the following reasons …'; semantics: 'I define the key concepts involved in my theory as …'; for similar questions see [59]). In step 2, opponents would try to connect their concrete scientific hypotheses or theories with the previously uncovered meta-scientific assumptions, imagining the former stemming from the latter. Step 3 involves requesting the rivals to write a short text tackling their points of agreement and disagreement, the strengths and shortcomings of the rival's proposal, and finally, why their perspective might constitute a better alternative to the scientific problem at hand, while briefly responding to the criticisms posed by the other camp. This short text would inform the drafting of the adversarial collaboration manuscript (see the main text).

Box 2. What adversarial collaboration is not

The emphasis on cooperation may lead some to misconstrue adversarial collaborations, particularly those dealing with conceptual issues. However, this approach to controversies is:

- Not a way to neutralize or to soften criticism. Finding common ground does not amount to encouraging agreement at all costs, or a reluctance to criticize rival proposals. Adversarial collaboration incentivizes criticism –even sharp and severe – but requires that it is scientifically sound and respectful: based on a fair portrayal of opposing claims, focused on addressing the rival's main concerns (instead of sidestepping them), and free from gratuitous derogatory labels.
- Not a substitute for replies and rejoinders. While some critical commentaries on scientific papers are not especially instructive [68], others are, particularly at the first stages of disagreements, where dissenters might feel the need to unconstrainedly express their criticisms. In that sense, adversarial collaboration might be best seen as a tool to prevent disagreements from becoming entrenched, substituting replies and rejoinders when there is the suspicion that scientists are starting to talk past each other.
- Not a way to hinder conceptual change. As the philosopher of science Thomas Kuhn pointed out [69], when new revolutionary ideas are put forward, these are rarely substantiated enough to be unanimously accepted by the scientific community. Some may worry that making proponents of new, disruptive ideas sit at a table with defenders of the status quo might press them to soften their boldest claims, thus weakening the potential for scientific change. However, proponents of a novel theory are not obliged to accept adversarial collaboration proposals. Instead, they may opt for the traditional ways to handle scientific controversies. Alternatively, if they were to accept adversarial collaboration invitations, they could argue that at least their work meets the important epistemic value of fruitfulness, stimulating debate.

about, for instance, the role of development in adaptive evolution can be resolved with scientific evidence only – at least in the short run [65]. The partial disconnection between facts and meta-scientific domains of disagreement demands a new kind of adversarial collaboration: theoretical adversarial collaboration (see [66] for an early precursor, and [67] for a candidate example in psychology).

The practicalities of theoretical adversarial collaboration differ only partially from those of its standard, empirical counterpart. Theoretical adversarial collaboration must also involve a bidirectional flow of input between opponents at all stages of the process. Likewise, it should include an impartial arbiter who oversees the interaction, and the end product should be a co-authored paper written following strict deadlines. Yet, it will also have its peculiarities (see Figure 1 for a three-step model; see also Box 2 for clarifications). Particularly, the central sections of the text, written by the adversaries, should be aimed at outlining their views and criticizing their rival's proposal, stressing points of agreement and departure, and emphasizing how their views connect to the above meta-scientific domains of scientific disagreement. Additionally, the arbiter's task in theoretical adversarial collaboration would involve writing the discussion, focused on summarizing the arguments provided and trying to construct a cogent narrative about where the conceptual disagreements really lie.

Concluding remarks

Scientific controversies can sharpen arguments, spur theory development, and inspire new empirical research. However, these positive outcomes are diminished when adversaries enter a confrontational mode, something which unfortunately happens more often than desired [29]. The scientific community must find ways to promote constructive disagreements. Adversarial collaboration can ensure that controversies are managed cooperatively, resulting in better arguments, sounder theories, and more relevant data [31]. Ultimately, this benefits the entire scientific community, which gains clarity and a more impartial assessment of the merits of each theory or hypothesis.

Here we have explored features of ecology and evolutionary biology that would at first sight make them less suited for adversarial collaborations, trying to provide some possible solutions. Now, what can ecologists and evolutionary biologists who want to engage in adversarial collaboration

Outstanding questions

How can one initiate an adversarial collaboration? Instead of writing a critical commentary, ask your rivals whether they are willing to cooperatively move the controversy forward.

How can suitable partners be found? Look for intellectually humble peers with whom you disagree but who seem genuinely curious about finding things out. Preferably avoid uncooperative, arrogant, or dogmatic colleagues.

How can a suitable mediator be found? Look for mutually respected figures with no vested theoretical interests in the controversy.

How can researchers implement adversarial collaborations? Agree to jointly devise a study or inform your research projects by this approach.

How can we promote adversarial collaboration at the community level? Launch dedicated sections in journals and give time for such kind of interaction at conferences.

How can we find out whether a disagreement is empirical or theoretical? Discuss whether a realistic empirical study could help advance the controversy (empirical). If it cannot, reflect on whether disagreements hinge on the meta-scientific domains described in the text (theoretical).

How can we find a suitable neutral organism for an adversarial collaboration? Survey species based on a collaborative literature review. Discuss with experts whether the candidate species are fitting.

What attitudes best promote successful adversarial collaborations? Intellectual humility (acknowledging that one may be wrong and that the opponent might have their point) and epistemic flexibility (openness to consider alternative viewpoints as long as they are scientifically sound).

How can one be a good adversarial collaborator? Back claims on sound reasons and evidence, avoid judgmental labels, and weigh your assertions on your degree of perceived certainty.

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do (see also Outstanding questions)? Beyond contacting rivals to collaborate in concrete studies, we suggest that applications for research projects be framed under an adversarial collaboration approach (e.g., allocating at least part of the resources to test competing hypotheses together with rivals). And what can scientific societies do? A possibility is that both conferences and journals incorporate sessions and sections for adversarial collaboration, where scientists can find time and space to engage with their opponents [70]. Journal editors and reviewers can also encourage opponents to launch an adversarial collaboration if they perceive that their disagreement has gotten stuck. More generally, increasing the demographic diversity of participants in scientific controversies and adversarial collaborations is also likely to result in more constructive interactions and more objective assessments [71]. All these measures are expected to bring tangible progress to many controversies in ecology and evolution, ultimately bringing the handling of disagreements closer to the scientific ethos of cooperatively seeking to understand how the world works.

What can be gained from adversarial collaboration? Universally accepted evidence, unexpected points of agreement, clarification of disagreement, and greater respect for scientific rivals.

Acknowledgments

The authors thank Ivan Prates for his careful reading of the first version of the manuscript, as well as the journal editor and the reviewers for their constructive criticism. Funding was provided by the Knut and Alice Wallenberg Foundation (KAW 2023.0241) to T.U.

Declaration of interests

The authors declare no competing interests.

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